

Loopring: Developing a faster and more affordable decentralized exchange protocol with Google Cloud

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ABOUT LOOPRING



About Loopring

Loopring choose scalability and speed. Google Cloud to improve transaction time zkRollup custom experience.

Loopring is one of the pioneers in the cryptocurrency world that is on a journey to develop faster and more affordable solutions for decentralized cryptocurrency exchanges. In March 2020, Loopring launched Loopring.io, the world's first cryptocurrency exchange platform based on their zkRollup technology, an open source technology that enables 1,000 times higher transaction throughput at one-thousandth of costs compared to other decentralized exchange methods. Loopring aims to build technologies and products to facilitate a more

Google Cloud results

- Compute Engine helps save 15% of computing costs with custom VM solutions, enabling a lower transaction fee for users
- Shortens deploy time from several minutes to seconds to improve employee efficiencies
- Enhances transaction computation performance to improve customer service

Shorter customer release time to market

secure and scalable cryptocurrency payment and trading platform for customers around the world.

Industries: Financial Services & Insurance

Location: China, US, Canada, and Belgium

Compute Engine

(<https://cloud.google.com/compute/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cryptocurrency has created one of the fastest growing and evolving financial markets in recent years. With more than 5,000 cryptocurrencies in

existence, we've seen a boom in the number of trading platforms since 2018. Today, there are more than 20,000 different types of trading platforms in the cryptocurrency space

(<https://www.marketsandmarkets.com/Market-Reports/cryptocurrency-market-158061641.html#:~:text=The%20cryptocurrency%20market%20was%20valued,6.18%25%20during%20the%20forecast%20period.>)

. Valued at over US\$550 billion, the market is projected to expand at an annual growth rate of 6.18% in the next five years.

Currently, most of the cryptocurrency trading services rely on centralized exchanges, which hold users' funds and trades in internal systems to execute transactions. Despite its convenience, centralized exchanges have been criticized for lacking transparency and exposing traders under high risk of losing their funds due to potential hacker attacks. To provide a truly transparent and secure mode of cryptocurrency trading, more and more crypto-transaction service providers around the world are beginning to explore decentralized exchanges. Among these companies is Loopring (<https://loopring.org/#/about>), one of the pioneers in cryptocurrency innovation that believes in providing excellent user experience while ensuring the highest level of transparency and security.

Launched in 2017, Loopring has always been focused on overcoming the two major disadvantages of decentralized cryptocurrency exchanges, which are lower efficiency and higher costs compared to centralized alternatives. In March 2020, Loopring launched Loopring.io, a

cryptocurrency exchange platform based on their zkRollup technology, which enables 1,000 times higher transaction throughput at one-thousandth of the cost, compared to other decentralized exchange methods. It does this by moving most of the transaction computation out of what is often an overloaded blockchain, which forms the backbone of cryptocurrency trading.

Six months after its launch, Loopring's exchange platform Loopring.io now has more than 7,000 users trading dozens of cryptocurrencies. To improve its service even further, Loopring moved a part of the transaction computation to Google Cloud (<https://cloud.google.com/>) in May 2020 to reduce customer waiting time and lower transaction fees.

"We have been evaluating the most popular cloud services to find the best ones to enhance our service," explains Daniel Wang, Loopring founder and CEO. "When we started running our transaction computation on Google Cloud, we realized that we were not only able to reduce costs, but also enhance the performance of the transaction processing mechanism."

"The custom
VMs on Google
Cloud allow us
to request

computing
resources
according to our
needs and save
15% of
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for the users of
our exchange
platform."

*—Daniel Wang, founder &
CEO, Loopring*

Leading the fast-changing crypto market with highly scalable and custom VMs

Agility is key in a fast-changing environment like the cryptocurrency market, which isn't bound by geographical borders. As a cloud-native company, Loopring believes in the flexibility and low operational costs of hosting its services in the cloud. This is why it is careful to choose the right cloud platform to build its service on a highly scalable infrastructure accessible to everyone around the world. After the launch of its exchange

platform this year, it decided to migrate its zero-knowledge prover, the most resource-demanding part of its transaction computation that verifies trade orders by generating proofs, to the virtual machines (VMs) on [Compute Engine](#) (/compute).

The ability to request for custom machines on Google Cloud is one of the primary reasons for Loopring's migration. As the amount of transactions that need to be processed varies from day to day, Loopring requires different combinations of CPU and internal memory for its virtual machines. With Google Cloud, it is able to customize its VMs and keep computing costs low. When the number of transactions increase, it can easily scale back up.

"The custom VMs on Google Cloud allows us to request computing resources according to our needs and save 15% of computing costs. This means we can minimize the transaction fees for the users of our exchange platform," says Wang.

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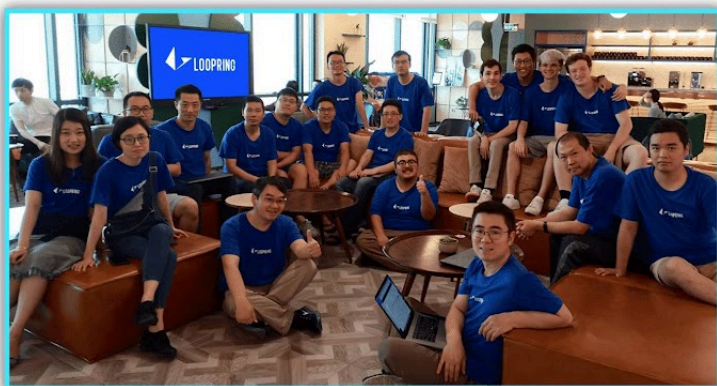
—Daniel Wang, founder & CEO, Loopring

Reducing customer wait time with faster zero-knowledge computation

The other reason Loopring switched to Google Cloud is the significantly faster computation speed supported by the N2 instance (/compute/docs/machine-types). In Loopring's unique transaction-processing mechanism, the time required to generate proofs via the zero-knowledge prover often causes a bottleneck in the system as it

is beyond the technical limitations of decentralized crypto exchanges to minimize withdrawal latency. As a solution, Loopring used to deploy its prover on another cloud provider, and it took around 10 minutes to generate proofs and complete the transaction, which meant that Loopring users had to wait for 10 minutes before they could receive their funds.

“The time required for proof computation directly impacts our user experience, because every second taken for proof generation is an extra second our users have to wait for their funds to be released,” details Wang. “With Google Cloud N2 instance, we’ve been able to reduce the computation time by 40%.”



Keeping a nimble team with an intuitive platform

Although Loopring still keeps its existing data on a previous cloud platform as backup, It has moved all of its proof-related data to Cloud Storage (/storage). “The amount of zero-knowledge proof data is huge, so transferring data between different cloud providers would incur considerable costs. Using the

same cloud platform saves us from extra spending and simplifies our infrastructure,” says Chao Ma, Chief Scientist at Loopring.

On the operations side, Loopring has noticed a big improvement in its ability to complete tasks quicker because of the speed of Google Cloud. Yan Yan, the engineer responsible for cloud operations at Loopring, noticed a significant difference in deploy time between Google Cloud and the previous cloud provider it used. “It used to take us several minutes to launch instances for an image of 40 G on the previous cloud platform. With Compute Engine, this same task only takes a few seconds. This has definitely helped boost our development and work efficiency,” she says.

Before using Google Cloud, the Loopring team frequently encountered configuration errors and had to spend a lot of time looking for mismatched parameters. But Yan Yan says that this rarely happens on Google Cloud because of the intuitive platform. Being more efficient at work with tools that are easy to operate gives the Loopring team more time to focus on research and development, which is key in this fast-moving environment.

Even though the company has 18 engineers today, only two of them are in charge of operations and maintenance. “We intend to remain small and nimble,” says Wang. “Google Cloud has helped us to do so by providing products with great performance, flexibility, and usability.”

“We've always
wanted to
deploy all our
services on the
same cloud
platform to
streamline
workflow, but
have been
concerned about
having to
change the
existing
infrastructure.
With Google
Kubernetes
Engine, we can
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knowing that it
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—Daniel Wang, founder &
CEO, Loopring

Improving transaction security and stability

As a cryptocurrency platform, security and accessibility is paramount to Loopring. To prevent distributed denial-of-service (DDoS) attacks, Loopring plans to explore the [DDoS protection solutions](/static/files/GCPDDoSprotection-04122016.pdf) (/static/files/GCPDDoSprotection-04122016.pdf) of Google Cloud soon to prepare for large-scale attacks in the future.

Since the rest of Loopring's transaction computation is already on Kubernetes, it plans to migrate fully to Google Cloud with [Google Kubernetes Engine](/kubernetes-engine) (/kubernetes-engine). "We've always wanted to deploy all our services on the same cloud platform to streamline workflow, but have been concerned about having to change the existing infrastructure. With Google Kubernetes Engine, we can rest assured knowing that it will be a seamless migration."

The support that the Google Cloud team provided has further sealed the team's thoughts on a full migration. "The help we've received from Google Cloud technical support and sales teams have made the transition to Google Cloud very easy for us. Prior to the migration, the Google Cloud team provided us with abundant resources to test its products, which has largely reduced the research and development cost of moving our services," says Wang.

"Migrating to a new cloud platform is a big decision to make. But the support that Google Cloud has provided smoothened the migration process and has given us confidence to make the leap," he

concludes. "Now that we have launched the Loopring Smart Wallet app (<https://loopring.io>), we are considering using Google Cloud as the preferred provider for our wallet back end."